

Groundwater in Two Lesser Himalayan Catchments of Himachal Pradesh, India: A Review of Resource, Quality and Recharge Studies

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Abstract

This paper reviews three groundwater studies from the Beas catchment in Himachal Pradesh. Two of them deal with the Suketi river basin in Mandi district: one describes the groundwater resource and proposes a development plan, and the other reports the chemistry of groundwater sampled in May and November 2009. The third study maps groundwater potential in the Gaj watershed of Kangra district for 2010 and 2022 using GIS and the analytic hierarchy process, and selects sites for recharge structures. In the Suketi basin, most of the usable groundwater lies in the alluvium of the Balh valley, where only about 22% of the assessed resource is in use. The hills depend on small springs, many of which almost stop flowing before the monsoon. The water is of Ca-HCO₃ type and is fit for drinking and irrigation, except for high fluoride (4.2 mg/L) in one bore well on granite and raised nitrate in two towns. In the Gaj watershed, the area of very high potential became smaller between 2010 and 2022 as monsoon rainfall weakened. The three studies cover quantity, quality and site selection separately, and none of them links all three. We point out the main gaps in data and method, and suggest a simple combined approach for future work.

Keywords: Groundwater, Springs, Water quality, Groundwater potential, Artificial recharge, Himachal Himalaya.

INTRODUCTION

Groundwater supplies a large share of the drinking and irrigation water used in India, and storage has been falling in many northern states (Rodell et al., 2009). In the hills of the Himalaya, the situation is different from the plains. Villages and small towns take their water from springs, shallow dug wells and hand-pump bore wells. These sources are fed by small aquifers in weathered and fractured rock, and steep slopes send much of the rain away as runoff. Several studies have reported falling spring discharge in the region (Valdiya and Bartarya, 1989; NITI Aayog, 2018).

Groundwater in the hills is also poorly measured. In Himachal Pradesh, official resource estimates are made only for valley areas, so the hill tracts, which cover most of the state, are not assessed (CGWB, 2006). At the same time, demand is rising in the foothill districts because of population growth, tourism and farming.

Three studies from the Beas catchment address parts of this problem. Pophare and Balpande (2015) described the groundwater resource of the Suketi river basin and proposed separate plans for its valley and its hills. Pophare et al. (2018) studied the chemistry of groundwater in the same basin. Sharma et al. (2025) mapped groundwater potential in the Gaj watershed and chose sites for artificial recharge. This review compares the three studies, brings their main results together and identifies what is still missing. Each paper was read in full, and its setting, data, methods, results and recommendations were noted. Values that we calculated from the published data are marked as derived.

STUDY AREAS AND METHODS

Table 1 summarises the two catchments and the three studies.

The Suketi basin has a central valley, the Balh valley, which holds up to about 370 m of Quaternary gravel, sand and clay. The valley lies between the Mandi thrust and the

Main Central Thrust, and is surrounded by hills of older meta-sedimentary rocks, granite and limestone (Srikantia and Bhargava, 1998). The Gaj watershed is larger, higher and much wetter. Its southern part has alluvium and Siwalik sediments, while the north and east are made of hard rocks. It is crossed by the Main Boundary Thrust and several other thrusts (Kumar and Mahajan, 2001). Both catchments therefore have one porous aquifer with good storage and a much larger area of fractured hard rock.

The three studies used very different methods. Pophare and Balpande (2015) relied on field work: they measured water levels in dug wells before and after the monsoon, collected tube-well records, visited springs and classed them by discharge after Meinzer (1923), and divided the basin into 53 small watersheds for planning. Pophare et al. (2018) analysed 14 parameters in each sample following APHA (1998) and judged the results against drinking-water limits and standard irrigation indices (Richards, 1954). Sharma et al. (2025) built eleven map layers, removed strongly related layers using the variance inflation factor, gave weights to the remaining layers with the analytic hierarchy process (Saaty, 1980) and combined them by weighted overlay. They made four maps (before and after the monsoon in 2010 and in 2022) by changing only the rainfall layer, and checked the maps with the area under the ROC curve.

RESULTS OF THE REVIEWED STUDIES

Where the Groundwater Is

Suketi valley

In the Balh valley, dug wells up to about 15 m deep tap an unconfined aquifer, and tube wells, mostly 50–125 m deep, tap layers of sand and gravel below clay (Pophare and Balpande, 2015). Some tube wells flow on their own at the surface. Most tube wells yield 9–45 L/s. In 2009 the water table was within 12 m of the surface before the monsoon

Table 1: Study areas and methods of the three reviewed studies.

	Pophare and Balpande (2015)	Pophare et al. (2018)	Sharma et al. (2025)
Catchment	Suketi river basin, Mandi district; $\sim 422 \text{ km}^2$; 760–2,890 m; rainfall $\sim 1,136 \text{ mm/yr}$		Gaj watershed, Kangra district; $\sim 900 \text{ km}^2$; up to $\sim 4,500 \text{ m}$; rainfall $\sim 2,600 \text{ mm/yr}$
Main aquifers	Balh valley fill (79 km^2); fractured meta-sediments, granite and limestone in hills		Southern alluvium and Siwalik units; Chail rocks and Dhauladhar granite
Aim	Assess the resource and plan its development	Assess water quality and its controls	Map groundwater potential and select recharge sites
Data	28 dug wells, 64 tube wells, 93 springs (2009); thematic maps	31 groundwater and 5 river samples, May and Nov. 2009	Satellite DEM and imagery, soil, rainfall and temperature data (2010, 2022)
Methods	Field inventory, water-level maps, morphometry, runoff estimates	Ion analysis, Piper and Gibbs plots, drinking and irrigation indices	Collinearity test, AHP weights, weighted overlay, ROC–AUC check

and within 10 m after it, and 27 of the 28 observation wells changed by less than 2 m between seasons. In more than half of the wells the water stood within 4 m of the surface.

The net annual groundwater available in the valley was estimated at 3.48 Mm^3 , while use was only 0.78 Mm^3 , so the stage of development was about 22% (CGWB, 2006). The valley aquifer is therefore under-used and could supply more water if pumping is planned with care.

Suketi hills

The hills are much poorer in groundwater. Most of the 93 springs studied are fracture or depression springs. In May 2009 many were dry or gave less than 0.01 L/s, and in November most gave only 0.1–0.4 L/s. On Meinzer's scale, most belong to the three smallest classes (VI to VIII). Only a few springs linked to limestone or major faults, near Nalni, Nagchalla and Mandi, gave more than 10 L/s. The drainage network is dense and fine, which means that rain leaves the slopes quickly (Pophare and Balpande, 2014). The authors identified three slope belts: hills steeper than 10° that mostly shed water, a foot-hill belt of $5\text{--}10^\circ$ that is the main recharge zone, and the flat valley floor where groundwater comes out.

Gaj watershed

In the Gaj watershed, zones of moderate to very high potential follow the main thrusts and lineaments and the southern alluvium, while the steep northern slopes score low (Sharma et al., 2025). Fig. 1 shows the share of the area in each class. The very high class fell from 4.20% to 2.63% of the area before the monsoon and from 4.46% to 3.20% after it. The maps scored AUC values of 0.82 and 0.81, which is usually read as good accuracy.

The two catchments point to the same controls even though the methods differ. Faults, thrusts and lineaments guide both spring locations and high-potential zones. Slope decides whether rain soaks in or runs off. Valley fill and alluvium are the only aquifers with large storage.

Water Quality

Only the Suketi basin has chemical data (Pophare et al., 2018). The main results are listed in Table 2. All waters are slightly alkaline and have low salt content. Conductivity

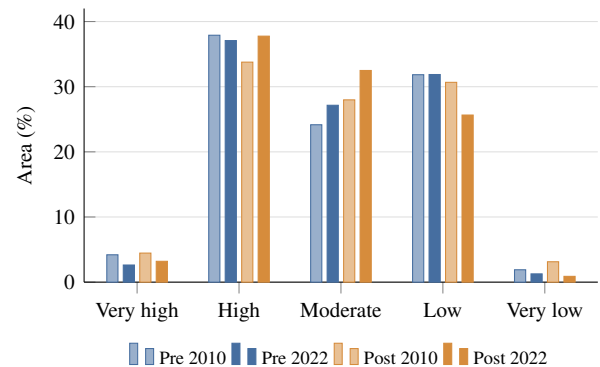


Figure 1: Groundwater potential classes in the Gaj watershed, drawn from the data of Sharma et al. (2025).

was highest in dug wells ($582 \mu\text{S/cm}$ in May) and lowest in springs ($350 \mu\text{S/cm}$), and every source was more dilute after the monsoon. Calcium and bicarbonate are the main ions, so most samples plot as Ca-HCO_3 water on the Piper diagram (Piper, 1944). On the Gibbs diagram most samples fall in the rainfall-dominated field (Gibbs, 1970). This suggests that water passes through the steep terrain quickly and has little time to react with the rock. Spring water and river water have almost the same ionic make-up, which indicates that hill springs feed the base flow of the Suketi river.

Two local problems were found. A bore well at Pabu, on granite, had 4.2 mg/L fluoride in May and 2.2 mg/L in November, above the limit of 1.5 mg/L (BIS, 2012; WHO, 2017). The authors linked this to fluoride released from the granite. Nitrate reached 36–37 mg/L in a spring at Mandi and 28–32 mg/L in a bore well at Rewalsar. These values are below the 45 mg/L limit, but both sites are in crowded towns, and nitrate is likely to come from sewage and farm runoff. For irrigation, all samples were safe on the sodium and carbonate indices. Six samples in each season were corrosive, which matters for metal pipes in supply schemes.

Change over Time

The Suketi data come from one year (2009), so they show only the seasonal pattern. The Gaj study covers 2010 to 2022. Over this period, monsoon rainfall in the watershed

Table 2: Main water-quality results for the Suketi basin, 2009 (Pophare et al., 2018).

Parameter	Limit	Result
pH	6.5–8.5	Mean 7.3–7.9
Water type	–	Mostly Ca–HCO ₃
Fluoride	1.5 mg/L	Exceeded in one bore well (4.2 mg/L)
Nitrate	45 mg/L	Within limit; 28–37 mg/L in two towns
SAR	<10	All samples
Sodium (%)	≤60	Excellent in 84–90% of samples
RSC	<1.25 meq/L	All samples
Corrosivity	<1	Six samples above 1 in each season

was lower in several years, and the area of very high potential decreased (Sharma et al., 2025). Government records quoted in that study show water-level falls of up to 7.16 m in the area, and 235 of the 4,535 hand pumps in the area needed repair or replacement, partly because of low water levels. Maximum temperature rose over the same period, and the population of Kangra district grew by about 12% between 2011 and 2023 (derived).

These trends should be read with care. Rainfall was the only layer that changed between the 2010 and 2022 maps, so the change in the maps reflects changed rainfall and not a measured change in groundwater storage.

Recharge and Management Proposals

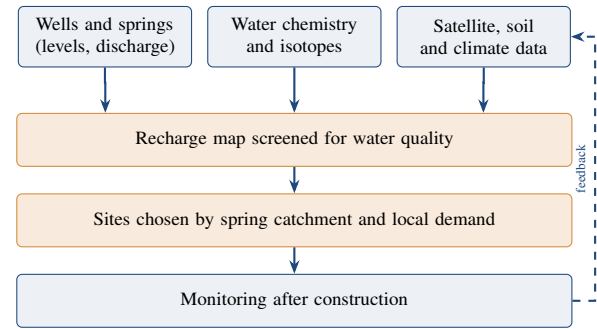
Pophare and Balpande (2015) suggested separate plans for the valley and the hills. In the valley, well-designed dug wells can meet most local demand, recharge should be tried where post-monsoon water levels are deeper than about 3 m, and old village ponds should be cleaned and deepened. In the hills, they proposed about three check dams or gabions for each spring, roughly 282 structures in all, together with rooftop rainwater harvesting and farm ponds. Structures were matched to stream order, from gully plugs on small streams to check dams on larger ones (CGWB, 2007). They also asked for the revival of traditional systems such as *baudis*, *nauns* and *kuhls* (Sharma and Kanwar, 2009).

Sharma et al. (2025) placed candidate sites where lineaments cross streams and kept only those on gentle or moderate slopes and suitable land. Priority was given to crowded foothill areas. They recommended check dams, gabions, contour trenches, pony-pit wells and roof-water tanks, and avoided steep slopes because of fast runoff and the risk of slope failure.

Both studies favour small, low-cost structures on small streams and gentle ground. Neither estimates how much water the structures would add, what they would cost, or who would maintain them. These points often decide whether recharge schemes work in practice (Bouwer, 2002).

DISCUSSION

The three studies are useful together because each covers a different question. The first shows where water is stored and how much is used. The second shows that the water is mostly safe and that hill springs feed the rivers. The third covers the whole watershed in a consistent way and looks at change over twelve years. Table 3 lists the main strengths

**Figure 2:** Suggested combined approach for groundwater planning in Lesser Himalayan catchments.

and weaknesses.

Three weaknesses are shared or especially important. First, the Suketi data are from a single year and are now old. Second, the Gaj maps have no direct field check. The authors compared them with an infiltration map, but that map was built from five layers that were also used in the potential model, so close agreement is partly expected. Testing against spring flow or well levels would be stronger (Machiwal et al., 2011). Third, no study looks at water quality and recharge together. The Suketi results show fluoride in granite and nitrate in towns. The Gaj study proposes recharge in crowded foothills and also has granite in its catchment, so recharge water there should be tested before structures are built.

The methods could also be shared between the two catchments. A spring and well inventory like the one in the Suketi basin would give the Gaj maps the field data they lack. In turn, the mapping method of the Gaj study could be used in the Suketi hills, where no official estimates exist, to decide which of the 53 small watersheds should get structures first.

Fig. 2 shows a simple combined approach. Field data, chemistry and satellite data are analysed separately and then merged into one recharge map that excludes polluted or unsuitable sites. Sites are planned at the scale of single spring catchments, as recommended for the Himalaya (NITI Aayog, 2018; Tambe et al., 2012), and a monitoring network feeds the results back into the next round of planning.

FUTURE WORK

1. Set up observation wells and gauged springs and measure them every season.
2. Use stable isotopes and geological mapping to find the recharge areas of important springs.
3. Extend water sampling to the Gaj watershed, and add trace metals and bacteria.
4. Test how sensitive the potential maps are to the chosen weights, and check them against field data.
5. Estimate how much water each recharge structure adds, and record its cost and upkeep.
6. Involve local communities, and restore traditional water systems along with new structures.

CONCLUSIONS

The studies reviewed here show that groundwater in these Himalayan catchments occurs in two very different settings. The valley fill of the Balh valley holds a large and little-used

Table 3: Strengths, limitations and main gaps of the reviewed studies.

	Pophare and Balpande (2015)	Pophare et al. (2018)	Sharma et al. (2025)
Strengths	Large field inventory of wells and springs; clear valley–hill contrast; planning by small watersheds	Both seasons and all source types sampled; full set of drinking and irrigation checks	Covers the whole watershed; compares two years; maps actual recharge sites
Limitations	One year of data; hills not quantified; number of structures based on a fixed rule	One year of data; no trace metals, bacteria or isotopes	No well or spring data; weights depend on expert opinion; check map shares inputs with the model
Main gap	Water budget of spring catchments	Long-term quality monitoring	Field validation and water-quality screening

resource with shallow water levels. The hills depend on small springs that weaken before every monsoon. The water of the Suketi basin is generally good for drinking and irrigation, but fluoride in granite areas and nitrate in towns need regular checks. In the Gaj watershed, the zones most favourable for groundwater shrank between 2010 and 2022 as rainfall declined.

Each study covers one part of the problem: quantity, quality or site selection. Future work should combine them, test maps against field data, screen recharge sites for water quality and plan at the scale of spring catchments. This would give local planners a sounder basis for managing groundwater in the Himalayan foothills.

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DECLARATIONS

Conflict of interest: The authors declare that they have no competing interests.

Data availability: All data discussed are taken from the published sources cited.

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